basis of the above considerations, the majority of physicists sum it up by saying there is a harmonious coexistence between special relativity and Aspect's results on entangled particles. In short, special relativity survives by the skin of its teeth. Many physicists find this convincing, but others have a nagging sense that there is more to the story.

At a gut level I've always shared the coexistence view, but there is no denying that the issue is delicate. At the end of the day, no matter what holistic words one uses or what lack of information one highlights, two widely separated particles, each of which is governed by the randomness of quantum mechanics, somehow stay sufficiently "in touch" so that whatever one does, the other instantly does too. And that seems to suggest that some kind of faster-than-light *something* is operating between them.

Where do we stand? There is no ironclad, universally accepted answer. Some physicists and philosophers have suggested that progress hinges on our recognizing that the focus of the discussion so far is somewhat misplaced: the real core of special relativity, they rightly point out, is not so much that light sets a speed limit, as that light's speed is something that all observers, regardless of their own motion, agree upon.[16] More generally, these researchers emphasize, the central principle of special relativity is that no observational vantage point is singled out over any other. Thus, they propose (and many agree) that if the egalitarian treatment of all constant-velocity observers could be squared with the experimental results on entangled particles, the tension with special relativity would be resolved.[17] But achieving this goal is not a trivial task. To see this concretely, let's think about how good old-fashioned textbook quantum mechanics explains the Aspect experiment.

According to standard quantum mechanics, when we perform a measurement and find a particle to be here, we cause its probability wave to change: the previous range of potential outcomes is reduced to the one actual result our measurement finds, as illustrated in Figure 4.7. Physicists say the measurement causes the probability wave to *collapse* and they envision that the larger the initial probability wave at some location, the larger the likelihood that the wave will collapse to that point —that is, the larger the likelihood that the particle will be found at that point. In the standard approach, the collapse happens instantaneously